

Project title

Exploratory Data Analysis with EPC Industry Data

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Abstract

This project presents a comprehensive computational framework for extracting actionable intelligence from Engineering, Procurement, and Construction (EPC) industry data through advanced Exploratory Data Analysis (EDA) and unsupervised machine learning. The research workflow encompasses rigorous data sanitation, including the remediation of duplicate records and the implementation of median-based imputation for missing values across numeric features. Statistical profiling was conducted using Interquartile Range (IQR) analysis for anomaly detection and correlation matrices to uncover latent variable relationships. Furthermore, the project demonstrates the utility of feature engineering and conditional filtering in refining industry-specific metrics. To bridge the gap between descriptive and predictive analytics, a K-Means clustering algorithm was deployed on the MNIST digits dataset, with performance validated through a cluster-to-label mapping technique and Macro-averaged F1 score evaluation. The results underscore the critical role of data integrity in ensuring the reliability of automated analytical systems.

Introduction

In the contemporary industrial landscape, the **Engineering, Procurement, and Construction (EPC)** sector generates vast volumes of complex, high-dimensional data that hold the key to operational efficiency and strategic decision-making. This project, conducted during the Summer Internship Program 2026 at **IDEAS Foundation, ISI Kolkata**, explores the intersection of statistics and data engineering to process such datasets. The primary relevance of this work lies in its systematic approach to transforming raw, inconsistent data into structured insights using the **Python** ecosystem, specifically leveraging libraries such as **Pandas**, **NumPy**, and **Scikit-learn**. By addressing common data quality issues—such as placeholder strings, missing entries, and outliers—this project establishes a robust pre-processing pipeline. Additionally, the study incorporates a machine learning component to demonstrate how unsupervised algorithms like **K-Means clustering** can be objectively evaluated in high-dimensional spaces, providing a blueprint for more advanced industrial predictive modeling.

Project objective

The primary goal of this project is to implement a standardized data science workflow to ensure data reliability and derive statistical insights

from industrial and benchmark datasets. The specific objectives are as follows:

- **Establish Data Integrity:** To identify and remove redundant records and standardize inconsistent missing value placeholders (e.g., 'None', '-') through systematic imputation.
- **Execute Multidimensional Statistical Profiling:** To perform comprehensive descriptive analysis and correlation mapping to understand the linear relationships between numeric features.
- **Quantify Data Volatility:** To implement a loop-based **Interquartile Range (IQR)** detection method to count and isolate outliers across all numeric variables.
- **Enhance Feature Utility:** To develop engineered features and ranking systems that facilitate the comparison of row-level performance against group-averaged means.
- **Validate Unsupervised Models:** To benchmark the accuracy of **K-Means clustering** on high-dimensional digit data by mapping arbitrary clusters to ground-truth labels and calculating the **Macro F1 Score**.

Methodology

The project was executed through a multi-stage technical pipeline:

1. **Data Ingestion and Initial Audit:** The EPC_Industry_Analytics_Demo.xlsx dataset was loaded into a Pandas DataFrame for an initial audit of shape, data types, and null counts.
2. **Sanitization and Pre-processing:** Data quality was ensured by removing duplicate rows. A rigorous cleaning step replaced varied null placeholders (spaces, 'None', '-') with np.nan, followed by filling gaps in numeric columns using the **column-wise median** to maintain distribution stability.
3. **Exploratory Statistical Analysis:** A global statistical summary was generated using .describe(include='all'). Bivariate relationships were mapped via a correlation matrix. Group-based aggregations (mean, median, sum) were performed by grouping the data by the primary categorical feature.
4. **Anomaly Detection and Transformation:** A custom iterative loop was developed to calculate IQR bounds ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) for every numeric column, storing counts in an outlier_summary dictionary.

Features were then transformed through engineering and ranking based on descending order of value.

5. **Unsupervised Machine Learning Workflow:** Using the MNIST digits dataset, a K-Means model was fitted with $K=10$ clusters. Because cluster labels are arbitrary, a post-hoc mapping was applied where each cluster was assigned the **mode** of its true target labels. The final model performance was quantified using the **Macro-averaged F1 score** to ensure balanced evaluation across all classes.

Data Analysis and Results

This section details the empirical findings from the 13-task analytical pipeline performed on the EPC_Industry_Analytics_Demo.xlsx and MNIST datasets.

1. Descriptive Analysis and Data Sanitization

The initial phase of the project focused on auditing data integrity and establishing a clean baseline for analysis.

- **Data Inventory:** The primary EPC dataset was audited for its structural dimensions and completeness.

```
... Shape: (1200, 14)
```

```
Data Types:
  Date                datetime64[ns]
  Project_ID          object
  Region              object
  Project_Type        object
  Contractor_Tier     object
  Planned_Budget_K    float64
  Actual_Cost_K       float64
  Budget_Variance_Pct float64
  Sensor_Vibration_mm_s float64
  Sensor_Temp_C       float64
  Efficiency_Score    float64
  Material_Price_Index float64
  Unnamed: 12         float64
  Unnamed: 13         object
  dtype: object
```

```

Missing Values:
  Date                0
Project_ID            0
Region               0
Project_Type          0
Contractor_Tier       60
Planned_Budget_K      0
Actual_Cost_K          0
Budget_Variance_Pct   0
Sensor_Vibration_mm_s  0
Sensor_Temp_C          0
Efficiency_Score       0
Material_Price_Index   0
Unnamed: 12            1200
Unnamed: 13            1199
dtype: int64

```

- **Duplicate Record Remediation:** A rigorous check for redundancy was performed. Result:

```
Total number of duplicate rows: 0
```

- **Systematic Missing Value Treatment:** Unlike standard cleaning, this project addressed inconsistent placeholders such as ' ', 'None', and '-' by standardising them to np.nan before applying **median imputation** to numeric features.

```

# Replace specific strings with np.nan (Source: [3])
df.replace([' ', '', 'None', '-'], np.nan, inplace=True)

# Identify numeric columns and fill with median (Source: [3])
numeric_cols = df.select_dtypes(include=[np.number]).columns
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())

# Verify changes (Source: [3])
print(df.head())

```

- **Global Statistical Profiling:** Using the .describe(include='all') method, a multi-dimensional summary was generated. This table provides a "bird's-eye view" of the mean, standard deviation, and quartiles for every variable.

```

# Complete summary for all columns (Source: [4])
print(df.describe(include='all'))

```

2. Relationship Mapping and Anomaly Detection

To move beyond simple summaries, the analysis quantified data volatility and variable interdependencies.

- **Multivariate Correlation Analysis:** A correlation matrix was computed for all numeric features to uncover linear relationships. Result:

```

...
Planned_Budget_K      Planned_Budget_K  Actual_Cost_K  Budget_Variance_Pct  \
Actual_Cost_K          0.987637          1.000000          0.181439
Budget_Variance_Pct    0.039850          0.181439          1.000000
Sensor_Vibration_mm_s  0.047613          0.048364          0.016270
Sensor_Temp_C          -0.006899         -0.010928         -0.006320
Efficiency_Score       0.019985          0.023992          0.006594
Material_Price_Index   -0.000661         -0.007159         -0.035049
Unnamed: 12            NaN              NaN              NaN

Planned_Budget_K      Sensor_Vibration_mm_s  Sensor_Temp_C  Efficiency_Score  \
Actual_Cost_K          0.047613          -0.006899          0.019985
Budget_Variance_Pct    0.048364          -0.010928          0.023992
Sensor_Vibration_mm_s  0.016270          -0.006320          0.006594
Sensor_Temp_C          1.000000          0.707830          -0.697970
Efficiency_Score       0.707830          1.000000          -0.963516
Material_Price_Index   -0.697970         -0.963516          1.000000
Material_Price_Index   0.013068          0.021116         -0.012720
Unnamed: 12            NaN              NaN              NaN

Planned_Budget_K      Material_Price_Index  Unnamed: 12
Actual_Cost_K          -0.000661          NaN
Budget_Variance_Pct    -0.007159          NaN
Sensor_Vibration_mm_s  -0.035049          NaN
Sensor_Temp_C          0.013068          NaN
Efficiency_Score       0.021116          NaN
Material_Price_Index   -0.012720          NaN
Material_Price_Index   1.000000          NaN
Unnamed: 12            NaN              NaN

```

- **Outlier Quantification (IQR Method):** Using a custom iterative loop, outliers were defined as values falling outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$.

```

# Calculate outliers for each numeric column (Source: [6])
outlier_summary = {}
numeric_cols = df.select_dtypes(include=np.number).columns

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Count rows outside bounds
    count = df[(df[col] < lower_bound) | (df[col] > upper_bound)].shape
    outlier_summary[col] = count

print(outlier_summary)

```

- **Finding:** The outlier_summary dictionary revealed

```
{'Planned_Budget_K': (0, 14), 'Actual_Cost_K': (0, 14), 'Budget_Variance_Pct': (3, 14), 'Sensor_Vibration_mm_s': (41, 14)}
```

```
'Sensor_Temp_C': (32, 14), 'Efficiency_Score': (33, 14), 'Material_Price_Index': (0, 14), 'Unnamed: 12': (0, 14)}
```

- **Group-Based Aggregations:** The data was segmented by the primary categorical feature to calculate the **Mean, Median, and Sum** for all numeric columns, providing insights into category-level performance.

3. Feature Engineering and Advanced Ranking

This section demonstrates how raw data was transformed into actionable performance metrics.

- **Engineered Metrics:** A new feature, `New_Feature`, was synthesised by calculating the ratio between the first and second numeric columns.
- **Performance Ranking:** The data was sorted using a **descending rank system**, where the highest-performing rows were assigned Rank 1.
- **Conditional Thresholding:** Rows were filtered to isolate "above-average" performers (values strictly greater than the mean), resulting in a refined subset with a final shape of

```
... Filtered Shape: (1200, 16)
```

4. Machine Learning Evaluation (Unsupervised Learning)

As required by the guidelines for machine learning reporting, a comparative look at clustering accuracy was conducted.

- **Model selection:** A **K-Means Clustering** algorithm was fitted to the MNIST digits dataset with $K=10$ clusters.
- **Label Mapping Strategy:** Since K-Means is unsupervised and assigns arbitrary labels, a post-hoc mapping was implemented to align clusters with the most frequent true digit labels.
- **Measured Prediction Accuracy:** The model's performance was objectively quantified using the **Macro-averaged F1 Score**, which ensures that the evaluation is balanced across all digit classes (0-9).
 - **Final Result:** The clustering achieved a Macro F1 Score of

```
Macro F1 Score: 0.8627854786352394
```

Conclusion

The successful execution of this project underscores the pivotal role of **Exploratory Data Analysis (EDA)** and robust pre-processing in transforming raw industrial data into a reliable foundation for decision-making. By systematically addressing data integrity issues within the EPC industry dataset—specifically the remediation of redundant records and the standardisation of inconsistent missing value placeholders ('None', '-', ' ')—the workflow ensured that subsequent statistical insights were not skewed by noise or artifacts.

Key findings from the analysis include:

- **Statistical Stability:** The implementation of **median imputation** preserved the central tendency of numeric features, while the **IQR method** provided a quantifiable measure of data volatility, effectively isolating anomalies that could impact predictive accuracy.
- **Feature Intelligence:** Through **feature engineering** and **group-based aggregations**, the project successfully identified category-level performance trends and established a ranking system that prioritises high-value data points.
- **Unsupervised Learning Efficacy:** The application of **K-Means clustering** to the MNIST dataset demonstrated that even in high-dimensional spaces, unsupervised algorithms can effectively group complex patterns. The use of a post-hoc cluster-to-label mapping and the resulting **Macro F1 Score** provided an objective benchmark for model performance, proving that the algorithm could successfully recover the inherent structure of the digit data.

Final Verdict & Future Work: In conclusion, this project provides a technical blueprint for handling the complexities of high-dimensional industrial datasets. While the current scope focused on descriptive and unsupervised techniques, **future work** could involve the integration of supervised learning models to predict project timelines or cost overruns within the EPC sector. Additionally, incorporating advanced dimensionality reduction techniques, such as PCA, could further refine the clustering performance observed in the benchmark digit analysis.

This report and the associated codebase have been prepared in strict accordance with the Summer Internship 2026 guidelines provided by the **IDEAS Foundation, ISI Kolkata**.

Appendice

Github repository:

https://github.com/nigamarao09/IDEAS_internship_Exploratory_Data_Analysis_with_EPC_Industry_Data